3.1 Database Tables

Table 3.1 ProjectForge snapshots

Column	Data type	Key / constraint	Description
project_id	TEXT	Primary key	Unique project identifier supplied by the application.
payload	TEXT	NOT NULL	Latest project state, context and build specification serialized as JSON.

Table 3.2 ProjectForge history

Column	Data type	Key / constraint	Description
id	INTEGER	Primary key; automatically assigned	Identifier of a saved project-state change.
project_id	TEXT	Nullable; no foreign-key constraint	Identifier of the associated project.
payload	TEXT	NOT NULL	Project snapshot serialized as JSON at the time of saving.

Table 3.3 ProjectForge events

Column	Data type	Key / constraint	Description
id	INTEGER	Primary key; AUTOINCREMENT	Global event sequence number used for ordered replay.
project_id	TEXT	NOT NULL; no foreign-key constraint	Identifier of the project associated with the event.
payload	TEXT	NOT NULL	Execution-event details serialized as JSON.

3.1 Database Tables Continued

Table 3.4 Generated NoteKeeper notes

Column	Data type	Key / constraint	Description
id	INTEGER	Primary key; NOT NULL; automatically assigned	Unique note identifier.
title	TEXT	NOT NULL; default empty string	Note title.
body	TEXT	NOT NULL; default empty string	Note content.

Table 3.5 ProjectForge indexes

Index	Table	Columns	Purpose
sqlite_autoindex_ snapshots_1	snapshots	project_id	Automatically enforces primary-key uniqueness.
events_project	events	project_id, id	Supports retrieval of a project's events in sequence.

Table 3.6 Logical project relationships

Parent	Related column	Cardinality	Enforcement
snapshots. project_id	history.project_id	One to many	Application-maintained relationship; no declared foreign key.
snapshots. project_id	events.project_id	One to many	Application-maintained relationship; no declared foreign key.

Table 3.7 Database scope

Item	Implemented scope
ProjectForge storage	SQLite file: backend/data/projects.sqlite3; application tables: snapshots, history and events.
Generated application storage	Separate SQLite file: generated NoteKeeper backend/notes.db; application table: notes.
Internal SQLite table	sqlite_sequence maintains AUTOINCREMENT counters; it is not an application entity.
Constraint interpretation	NOT NULL rejects NULL but does not reject empty strings. Logical project relationships are not database-enforced foreign keys.